

RAILWAY RESERVATION SYSTEM

Use Case Flow – Booking, Cancellation and Administrator

Project Title: Railway Reservation System with Booking and Cancellation Features

Use Case Flow 1: Booking a Ticket

Goal	Book Railway Ticket
Actor	Passenger with help from other actors
Main flow	1. Passenger selects a train booking option. 2. The system requests the passenger to log in. 3. The passenger provides valid credentials. 4. The system invokes the Authentication service to verify the credentials. 5. Upon successful authentication, the passenger proceeds with the booking. 6. The system displays available trains based on source, destination, and journey date. 7. The passenger selects a train and preferred travel class. 8. The system checks seat availability for the selected train and class. 9. The passenger enters the required passenger details. 10. The system validates the passenger details and calculates the total fare. 11. The system displays the booking summary and payment options. 12. The passenger selects a payment method and makes the payment. 13. The Payment Gateway authorizes the payment. 14. Upon successful payment, the system confirms the ticket booking. 15. The system generates a unique PNR/Booking ID and updates the available seat count. 16. The system displays the ticket confirmation and booking details. 17. Ticket booking process is completed.
Alternate Flow	Invalid Login Credentials At Step 4, if authentication fails: 4a. System displays an error message and asks the passenger to retry. 4b. Passenger may retry login or cancel the booking. No Seats Available At Step 8, if no seats are available: 8a. System displays a "No seats available" message. 8b. Passenger may select another train/class or modify the search. Invalid Passenger Details At Step 10, if passenger details are invalid: 10a. System displays an appropriate validation error. 10b. Passenger corrects the details and continues. Payment Failure At Step 13, if payment fails: 13a. System displays a payment failure message. 13b. Passenger may retry payment or cancel the booking. 13c. Ticket should not be confirmed unless payment is successfully completed.

Use Case Flow 2: Cancelling a Ticket

Goal	Cancel Railway Ticket
Actor	Passenger with help from other actors
Main flow	1. Passenger selects the ticket cancellation option. 2. The system requests the passenger to log in. 3. The passenger provides valid credentials. 4. The system invokes the Authentication service to verify the credentials. 5. Upon successful authentication, the passenger proceeds with cancellation. 6. The system displays the passenger's booked tickets. 7. The passenger selects the ticket/PNR to be cancelled. 8. The system verifies the PNR and booking status. 9. The system checks whether the selected ticket is eligible for cancellation. 10. The system calculates the applicable cancellation charge and refund amount. 11. The system displays the cancellation details and refund amount. 12. The passenger confirms the cancellation. 13. The system updates the ticket status to Cancelled. 14. The system updates the seat availability for the cancelled ticket. 15. The system initiates the applicable refund through the Payment Gateway. 16. The system displays the cancellation confirmation and refund details. 17. Ticket cancellation process is completed.
Alternate Flow	Invalid Login Credentials At Step 4, if authentication fails: 4a. System displays an error message and asks the passenger to retry. 4b. Passenger may retry login or cancel the operation. Invalid PNR At Step 8, if the PNR is invalid or does not exist: 8a. System displays an "Invalid PNR" message. 8b. Passenger may enter another PNR or cancel the operation. Ticket Not Eligible for Cancellation At Step 9, if the ticket cannot be cancelled: 9a. System displays a message explaining that cancellation is not allowed according to the applicable policy. 9b. The ticket remains active. Passenger Cancels the Operation At Step 12, if the passenger selects No/Cancel: 12a. System stops the cancellation process. 12b. The ticket remains confirmed/active. Refund Failure At Step 15, if the refund cannot be processed immediately: 15a. System records the refund status as pending/failed according to the implementation. 15b. System displays the applicable refund status to the passenger.

Use Case Flow 3: Administrator Management

Goal	Manage Railway Reservation System
Actor	Administrator with help from other system services
Main flow	- Administrator opens the Railway Reservation System. - The system requests the administrator to log in. - The administrator provides valid administrator credentials. - The system invokes the Authentication service to verify the administrator credentials. - Upon successful authentication, the system displays the Administrator Dashboard. - The administrator selects a management option such as Trains, Stations, Routes, Schedules, Seats, Bookings, or Cancellations. - The system displays the relevant records and available management operations. - The administrator selects the required operation, such as Add, Update, Delete, or View. - The administrator enters or modifies the required railway information. - The system validates the entered information. - The system saves the valid changes to the database. - The system updates the affected train, route, schedule, seat, booking, or cancellation records. - The system displays a confirmation message for the completed operation. - The administrator can perform another management operation or log out. - Administrator management process is completed.
Alternate Flow	Invalid Administrator Credentials At Step 4, if authentication fails: - System displays an error message and asks the administrator to retry. - Administrator may retry login or exit the system. Invalid Train / Station / Route Details At Step 10, if entered information is invalid or incomplete: - System displays an appropriate validation error. - Administrator corrects the information and submits it again. Duplicate Train Number At Step 11, if the train number already exists: - System rejects the new train record. - System displays a message such as "Train number already exists." - Administrator enters a different train number. Record Not Found At Step 8, if the selected train, route, schedule, or other record does not exist: - System displays a "Record not found" message. - Administrator may return to the management screen and select another record. Delete Operation Cancelled At Step 8, if the administrator chooses Delete but cancels the confirmation: - System does not delete the record. - The administrator is returned to the management screen. Unauthorized Operation At Step 8, if the administrator attempts an operation for which access is not permitted: - System denies the operation. - System displays an appropriate authorization message.

